

CHAPTER 1: SOME BASIC CONCEPTS OF CHEMISTRY (JEE FOCUS)

Topics Covered:

- Nature of matter
- Properties of matter (mass, volume, density)
- Laws of chemical combination
- Dalton's atomic theory
- Atomic mass & molecular mass
- Mole concept & Avogadro number
- Molar mass & molar volume
- Percentage composition
- Empirical formula & molecular formula
- Stoichiometry & limiting reagent

Core Concepts:

- Matter → has mass + occupies space
- Density = mass/volume
- Conservation of mass → total mass constant
- Definite proportions → fixed mass ratio
- Multiple proportions → simple whole number ratios
- Atomic mass → relative to 1/12 of C-12
- Molecular mass → sum of atomic masses
- Mole → 6.022×10^{23} entities
- Molar mass → mass of 1 mole (g/mol)
- Avogadro number links particles ↔ moles
- Molar volume (gas, STP) → 22.4 L/mol
- % composition → fraction of element in compound
- Empirical formula → simplest ratio
- Molecular formula → actual ratio
- Stoichiometry → quantitative relation in reaction
- Limiting reagent → reactant consumed first
- Excess reagent → remains after reaction

Key Formulas:

- Mole = given mass / molar mass
- Particles = moles $\times$ N_A
- $N_A = 6.022 \times 10^{23}$
- Density = mass/volume
- % composition = (mass of element / molar mass) $\times$ 100
- Empirical formula → mole ratio (simplest)
- n = molecular mass / empirical mass
- Molecular formula = (Empirical formula)ⁿ
- Gas volume (STP) = moles $\times$ 22.4 L

Important Patterns:

- Always convert → mass → moles → ratio
- Assume 100 g in % composition problems
- Divide by smallest mole value for ratios

- Decimal ratio → multiply to nearest integer
- Limiting reagent → compare mole ratio, not mass
- Balanced equation → mandatory before solving
- Gas problems → check STP condition
- Units consistency → g, mol, L alignment

Common Mistakes:

- Direct mass ratio instead of mole ratio
- Skipping balancing of equation
- Ignoring limiting reagent
- Wrong molar mass calculation
- Not converting units properly
- Rounding empirical ratios incorrectly
- Confusing empirical vs molecular formula
- Using 22.4 L when not at STP

Examples (Important Cases):

- % composition of H₂O, CO₂
- Empirical formula from % data (CH₂O type)
- Molecular formula using molar mass
- Limiting reagent in H₂ + O₂ → H₂O
- Mole ↔ particles conversion

Exam Tips:

- Start with given → find moles → apply ratio
- Write units at every step
- Memorize atomic masses (H, C, O, N)
- Use shortcut → 1 mole gas = 22.4 L (STP only)
- Practice limiting reagent (high weight)
- Check final answer logically (sanity check)
- Avoid skipping intermediate steps